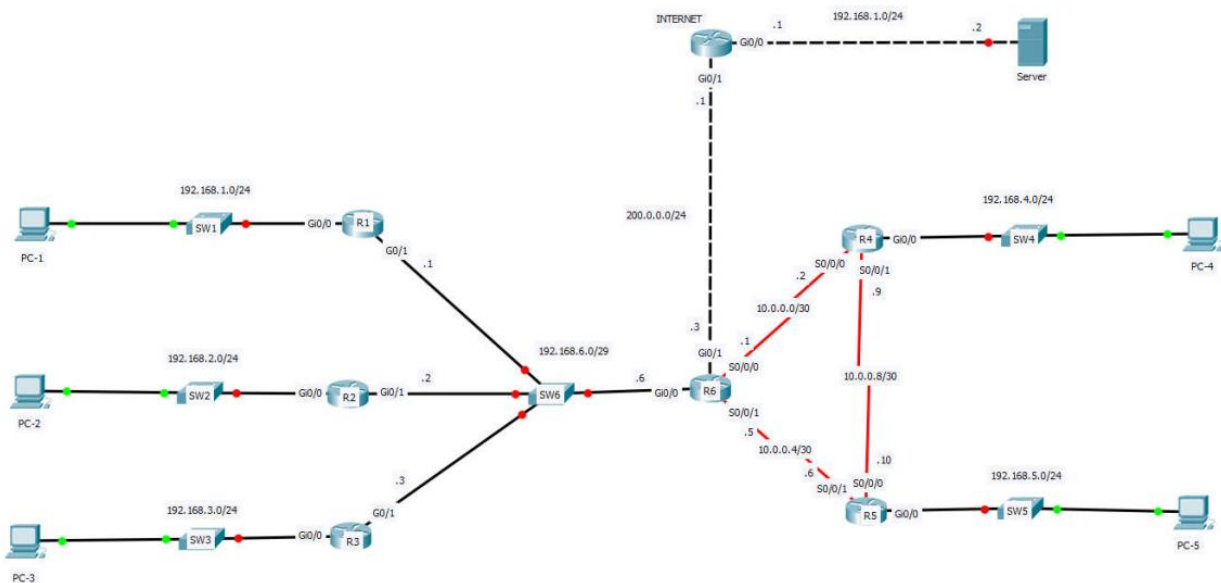


HƯỚNG DẪN LAB

LAB 9 – Lab tổng hợp 1

Designed by : Nguyễn Phú Thịnh

Sơ đồ LAB :



Yêu cầu :

- Cấu nối và đặt IP như sơ đồ
- Cấu hình SSH cho tất cả Router
- Cấu hình Router R6 làm DHCP Server cho các subnet 192.168.1.0/24, 192.168.2.0/24 và 192.168.3.0/24
- Cấu hình Router R4 và R5 lần lượt làm DHCP Server cho subnet 192.168.4.0/24 và 192.168.5.0/24
- Cấu hình định tuyến để các PC trao đổi dữ liệu được với nhau và các PC có thể truy cập Internet
- Cấu hình NAT để các PC có thể truy cập Internet
- Cấu hình NAT tĩnh trên router Internet để hosting cho Server
- Cấu hình Access List chặn các PC trong mạng 192.168.4.0/24 và 192.168.5.0/24 không thể download file bằng FTP và TFTP trên Server

Các bước thực hiện :

Bước 1: Đấu nối như sơ đồ.

Bước 2: Đặt tên và IP cho các router

R1

```
Router(config)# hostname R1
R1(config)# enable secret N3w$t@r
R1(config)# interface Gi0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shut
R1(config-if)# exit
R1(config)# interface Gi0/1
R1(config-if)# ip address 192.168.6.1 255.255.255.248
R1(config-if)# no shutdown
```

Kiểm tra :

```
R1# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	192.168.6.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

R2

```
Router(config)# hostname R2
R2(config)# enable secret N3w$t@r
R2(config)# interface Gi0/0
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit
R2(config)# interface Gi0/1
R2(config-if)# ip address 192.168.6.2 255.255.255.248
R2(config-if)# no shutdown
```

Kiểm tra:

```
R2# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.2.1	YES	manual	up	up
GigabitEthernet0/1	192.168.6.2	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

R3

```
Router(config)# hostname R3
R3(config)# enable secret N3w$t@r
R3(config)# interface Gi0/0
R3(config-if)# ip address 192.168.3.1 255.255.255.0
R3(config-if)# no shutdown
R3(config-if)# exit
R3(config)# interface Gi0/1
R3(config-if)# ip address 192.168.6.3 255.255.255.248
R3(config-if)# no shutdown
```

Kiểm tra :

```
R3# show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.3.1	YES	manual	up	up
GigabitEthernet0/1	192.168.6.3	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

R4

```
Router(config)# hostname R4
R4(config)# enable secret N3w$t@r
R4(config)# interface Gi0/0
R4(config-if)# ip address 192.168.4.1 255.255.255.0
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)# interface S0/0/0
R4(config-if)# ip address 10.0.0.2 255.255.255.252
R4(config-if)# no shutdown
R4(config-if)# exit
R4(config)# interface S0/0/1
R4(config-if)# ip address 10.0.0.9 255.255.255.252
R4(config-if)# no shutdown
```

Kiểm tra :

R4# show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.4.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	10.0.0.2	YES	manual	down	down
Serial0/0/1	10.0.0.9	YES	manual	down	down
Vlan1	unassigned	YES	unset	administratively down	down

R5

```
Router(config)# hostname R5
R5(config)# enable secret N3w$t@r
R5(config)# interface Gi0/0
R5(config-if)# ip address 192.168.5.1 255.255.255.0
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)# interface S0/0/0
R5(config-if)# ip address 10.0.0.10 255.255.255.252
R5(config-if)# no shutdown
R5(config-if)# exit
R5(config)# interface S0/0/1
R5(config-if)# ip address 10.0.0.6 255.255.255.252
R5(config-if)# no shutdown
```

Kiểm tra :

R5# show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.5.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	10.0.0.10	YES	manual	up	up
Serial0/0/1	10.0.0.6	YES	manual	down	down
Vlan1	unassigned	YES	unset	administratively down	down

R6					
<pre>Router(config)# hostname R6 R6(config)# enable secret N3w\$t@r R6(config)# interface Gi0/0 R6(config-if)# ip address 192.168.6.6 255.255.255.248 R6(config-if)# no shutdown R6(config-if)# exit R6(config)# interface Gi0/1 R6(config-if)# ip address 200.0.0.3 255.255.255.0 R6(config-if)# no shutdown R6(config-if)# exit R6(config)# interface S0/0/0 R6(config-if)# ip address 10.0.0.1 255.255.255.252 R6(config-if)# no shutdown R6(config-if)# exit R6(config)# interface S0/0/1 R6(config-if)# ip address 10.0.0.5 255.255.255.252 R6(config-if)# no shutdown</pre>					
Kiểm tra :					
R6# show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.6.6	YES	manual	up	up
GigabitEthernet0/1	200.0.0.3	YES	manual	up	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Serial0/0/0	10.0.0.1	YES	manual	up	up
Serial0/0/1	10.0.0.5	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

INTERNET					
<pre>Router(config)# hostname INTERNET INTERNET(config)# enable secret N3w\$t@r INTERNET(config)# interface Gi0/0 INTERNET(config-if)# ip address 192.168.1.1 255.255.255.0 INTERNET(config-if)# no shutdown INTERNET(config-if)# exit INTERNET(config)# interface Gi0/1 INTERNET(config-if)# ip address 200.0.0.1 255.255.255.0 INTERNET(config-if)# no shut</pre>					
Kiểm tra :					
INTERNET# show ip interface brief					
Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	200.0.0.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Bước 3 : Cấu hình SSH cho các router

R1					
<pre>R1(config)# username admin password N3w\$t@r R1(config)# ip domain-name example.com R1(config)# crypto key genera te rsa The name for the keys will be: R1.example.com Choose the size of the key modulus in the range of 360 to 2048 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.</pre>					

```
How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R1(config)# line vty 0 4
R1(config-line)# login local
R1(config-line)# transport input ssh
```

R2

```
R2(config)# username admin password N3w$t@r
R2(config)# ip domain-name example.com
R2(config)# crypto key genera te rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R2(config)# line vty 0 4
R2(config-line)# login local
R2(config-line)# transport input ssh
```

R3

```
R3(config)# username admin password N3w$t@r
R3(config)# ip domain-name example.com
R3(config)# crypto key genera te rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R3(config)# line vty 0 4
R3(config-line)# login local
R3(config-line)# transport input ssh
```

R4

```
R4(config)# username admin password N3w$t@r
R4(config)# ip domain-name example.com
R4(config)# crypto key genera te rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R4(config)# line vty 0 4
R4(config-line)# login local
R4(config-line)# transport input ssh
```

R5

```
R5(config)# username admin password N3w$t@r
R5(config)# ip domain-name example.com
R5(config)# crypto key genera te rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R5(config)# line vty 0 4
R5(config-line)# login local
R5(config-line)# transport input ssh
```

R6

```
R6(config)# username admin password N3w$t@r
R6(config)# ip domain-name example.com
R6(config)# crypto key genera te rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R6(config)# line vty 0 4
R6(config-line)# login local
R6(config-line)# transport input ssh
```

INTERNET

```
INTERNET(config)# username admin password N3w$t@r
INTERNET(config)# ip domain-name example.com
INTERNET(config)# crypto key genera te rsa
The name for the keys will be: R1.example.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

INTERNET(config)# line vty 0 4
INTERNET(config-line)# login local
INTERNET config-line)# transport input ssh
```

Bước 4 : Cấu hình Router 6 làm DHCP Server cho subnet 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24

R6
R6(config)# ip dhcp pool LAN-1 R6(dhcp-config)# network 192.168.1.0 255.255.255.0 R6(dhcp-config)# default-router 192.168.1.1 R6(dhcp-config)# dns 8.8.8.8 R6(dhcp-config)# exit R6(config)# ip dhcp pool LAN-2 R6(dhcp-config)# network 192.168.2.0 255.255.255.0 R6(dhcp-config)# default-router 192.168.2.1 R6(dhcp-config)# dns 8.8.8.8 R6(dhcp-config)# exit R6(config)# ip dhcp pool LAN-3 R6(dhcp-config)# network 192.168.3.0 255.255.255.0 R6(dhcp-config)# default-router 192.168.3.1 R6(dhcp-config)# dns 8.8.8.8 R6(dhcp-config)# exit R6(config)# ip dhcp excluded-address 192.168.1.1 R6(config)# ip dhcp excluded-address 192.168.2.1 R6(config)# ip dhcp excluded-address 192.168.3.1 R6(config)# ip route 192.168.1.0 255.255.255.0 192.168.6.1 R6(config)# ip route 192.168.2.0 255.255.255.0 192.168.6.2 R6(config)# ip route 192.168.3.0 255.255.255.0 192.168.6.3

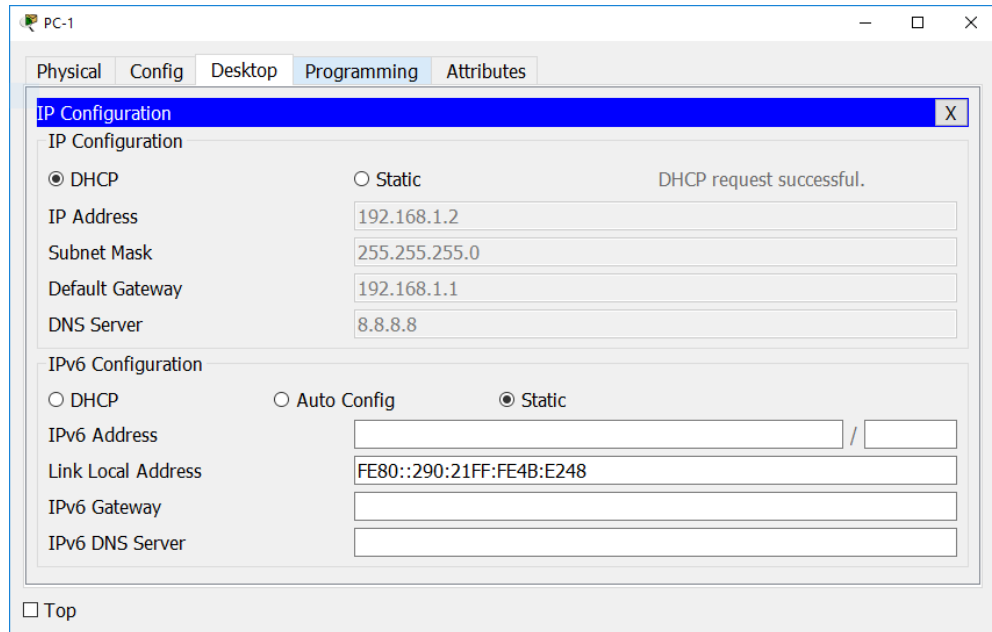
R1
R1(config)# interface Gi0/0 R1(config-if)# ip helper-address 192.168.6.6

R2
R2(config)# interface Gi0/0 R2(config-if)# ip helper-address 192.168.6.6

R3
R3(config)# interface Gi0/0 R3(config-if)# ip helper-address 192.168.6.6

Kiểm tra : lần lượt cho các PC-1, PC-2 và PC-3 nhận IP bằng DHCP

PC-1 :

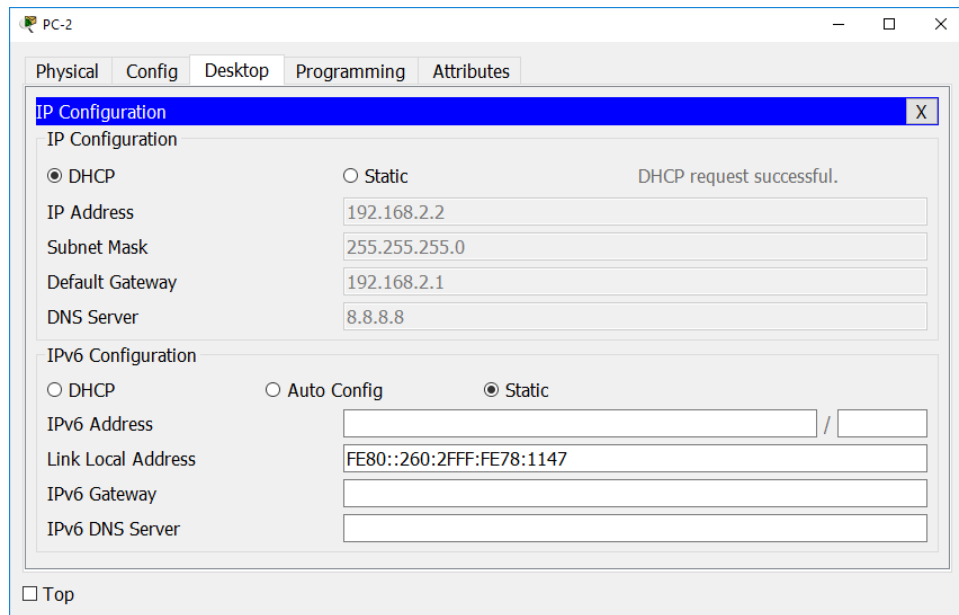


The screenshot shows the 'IP Configuration' window for PC-1. The 'Config' tab is selected. Under 'IP Configuration', the 'DHCP' radio button is selected, and a message 'DHCP request successful.' is displayed. The IP Address is 192.168.1.2, Subnet Mask is 255.255.255.0, Default Gateway is 192.168.1.1, and DNS Server is 8.8.8.8. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address field is empty, and the Link Local Address is FE80::290:21FF:FE4B:E248. The IPv6 Gateway and IPv6 DNS Server fields are also empty. A 'Top' button is at the bottom left.

IP Configuration	
☒ DHCP ☐ Static DHCP request successful.	
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	8.8.8.8
IPv6 Configuration	
☐ DHCP ☐ Auto Config ☒ Static	
IPv6 Address	
Link Local Address	FE80::290:21FF:FE4B:E248
IPv6 Gateway	
IPv6 DNS Server	

☐ Top

PC-2 :

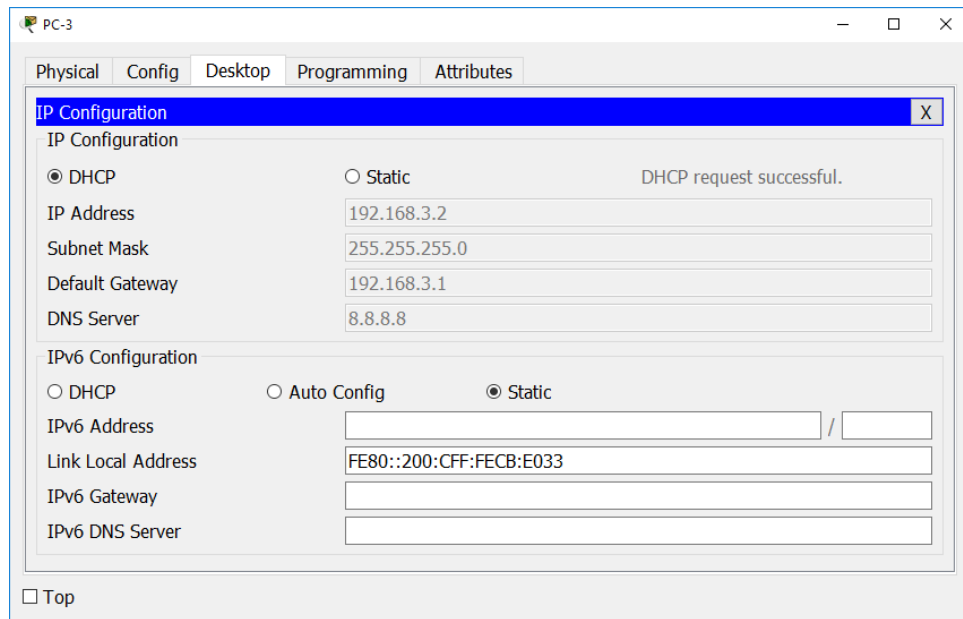


The screenshot shows the 'IP Configuration' window for PC-2. The 'Config' tab is selected. Under 'IP Configuration', the 'DHCP' radio button is selected, and a message 'DHCP request successful.' is displayed. The IP Address is 192.168.2.2, Subnet Mask is 255.255.255.0, Default Gateway is 192.168.2.1, and DNS Server is 8.8.8.8. Under 'IPv6 Configuration', the 'Static' radio button is selected. The IPv6 Address field is empty, and the Link Local Address is FE80::260:2FFF:FE78:1147. The IPv6 Gateway and IPv6 DNS Server fields are also empty. A 'Top' button is at the bottom left.

IP Configuration	
☒ DHCP ☐ Static DHCP request successful.	
IP Address	192.168.2.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	8.8.8.8
IPv6 Configuration	
☐ DHCP ☐ Auto Config ☒ Static	
IPv6 Address	
Link Local Address	FE80::260:2FFF:FE78:1147
IPv6 Gateway	
IPv6 DNS Server	

☐ Top

PC-3 :



Bước 5 : Cấu hình DHCP Server trên R4 và R5

R4
R4(config)# ip dhcp pool LAN-4
R4(dhcp-config)# network 192.168.4.0 255.255.255.0
R4(dhcp-config)# default-router 192.168.4.1
R4(dhcp-config)# dns 8.8.8.8
R4(dhcp-config)# exit
R4(config)# ip dhcp excluded-address 192.168.4.1

R5
R5(config)# ip dhcp pool LAN-5
R5(dhcp-config)# network 192.168.5.0 255.255.255.0
R5(dhcp-config)# default-router 192.168.5.1
R5(dhcp-config)# dns 8.8.8.8
R5(dhcp-config)# exit
R5(config)# ip dhcp excluded-address 192.168.5.1

Kiểm tra : lần lượt cho các PC-4, PC-5 nhận IP bằng DHCP

PC-4 :

The screenshot shows the configuration window for PC-4. The 'Config' tab is selected. The 'IP Configuration' section is active, showing 'DHCP' as the selected method. The IP Address is 192.168.4.2, Subnet Mask is 255.255.255.0, Default Gateway is 192.168.4.1, and DNS Server is 8.8.8.8. The 'IPv6 Configuration' section is also visible, with 'Static' selected. The 'Link Local Address' is FE80::202:16FF:FE72:9A86. A 'Top' button is at the bottom left.

IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IP Address	192.168.4.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	8.8.8.8

IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	/
Link Local Address	FE80::202:16FF:FE72:9A86
IPv6 Gateway	
IPv6 DNS Server	

☐ Top

PC-5 :

The screenshot shows the configuration window for PC-5. The 'Config' tab is selected. The 'IP Configuration' section is active, showing 'DHCP' as the selected method. The IP Address is 192.168.5.2, Subnet Mask is 255.255.255.0, Default Gateway is 192.168.5.1, and DNS Server is 8.8.8.8. The 'IPv6 Configuration' section is also visible, with 'Static' selected. The 'Link Local Address' is FE80::202:4AFF:FE11:C060. A 'Top' button is at the bottom left.

IP Configuration	
☒ DHCP	☐ Static
DHCP request successful.	
IP Address	192.168.5.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.5.1
DNS Server	8.8.8.8

IPv6 Configuration	
☐ DHCP	☐ Auto Config
☒ Static	
IPv6 Address	/
Link Local Address	FE80::202:4AFF:FE11:C060
IPv6 Gateway	
IPv6 DNS Server	

☐ Top

Bước 5 : Cấu hình định tuyến

R1	
R1(config)#	ip route 192.168.2.0 255.255.255.0 192.168.6.2
R1(config)#	ip route 192.168.3.0 255.255.255.0 192.168.6.3
R1(config)#	ip route 0.0.0.0 0.0.0.0 192.168.6.6
Kiểm tra :	
R1#	show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP	
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area	
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2	
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP	
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area	
* - candidate default, U - per-user static route, o - ODR	
P - periodic downloaded static route	
Gateway of last resort is 192.168.6.6 to network 0.0.0.0	
	192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.1.0/24 is directly connected, GigabitEthernet0/0
L	192.168.1.1/32 is directly connected, GigabitEthernet0/0
S	192.168.2.0/24 [1/0] via 192.168.6.2
S	192.168.3.0/24 [1/0] via 192.168.6.3
	192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.6.0/29 is directly connected, GigabitEthernet0/1
L	192.168.6.1/32 is directly connected, GigabitEthernet0/1
S*	0.0.0.0/0 [1/0] via 192.168.6.6

R2	
R2(config)#	ip route 192.168.1.0 255.255.255.0 192.168.6.1
R2(config)#	ip route 192.168.3.0 255.255.255.0 192.168.6.3
R2(config)#	ip route 0.0.0.0 0.0.0.0 192.168.6.6
Kiểm tra :	
R2#	show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP	
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area	
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2	
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP	
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area	
* - candidate default, U - per-user static route, o - ODR	
P - periodic downloaded static route	
Gateway of last resort is 192.168.6.6 to network 0.0.0.0	
S	192.168.1.0/24 [1/0] via 192.168.6.1
	192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.2.0/24 is directly connected, GigabitEthernet0/0
L	192.168.2.1/32 is directly connected, GigabitEthernet0/0
S	192.168.3.0/24 [1/0] via 192.168.6.3
	192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.6.0/29 is directly connected, GigabitEthernet0/1
L	192.168.6.2/32 is directly connected, GigabitEthernet0/1
S*	0.0.0.0/0 [1/0] via 192.168.6.6

R3

```
R3(config)# ip route 192.168.1.0 255.255.255.0 192.168.6.1
R3(config)# ip route 192.168.2.0 255.255.255.0 192.168.6.2
R3(config)# ip route 0.0.0.0 0.0.0.0 192.168.6.6
```

R3# **show ip route**

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 192.168.6.6 to network 0.0.0.0

```
S    192.168.1.0/24 [1/0] via 192.168.6.1
S    192.168.2.0/24 [1/0] via 192.168.6.2
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, GigabitEthernet0/0
L    192.168.3.1/32 is directly connected, GigabitEthernet0/0
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/29 is directly connected, GigabitEthernet0/1
L    192.168.6.3/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 192.168.6.6
```

R4

```
R4(config)# ip route 192.168.5.0 255.255.255.0 10.0.0.10
R4(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Kiểm tra :

R4# **show ip route**

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.0.0.1 to network 0.0.0.0

```
    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.0.0/30 is directly connected, Serial0/0/0
L    10.0.0.2/32 is directly connected, Serial0/0/0
C    10.0.0.8/30 is directly connected, Serial0/0/1
L    10.0.0.9/32 is directly connected, Serial0/0/1
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/24 is directly connected, GigabitEthernet0/0
L    192.168.4.1/32 is directly connected, GigabitEthernet0/0
S    192.168.5.0/24 [1/0] via 10.0.0.10
S*   0.0.0.0/0 [1/0] via 10.0.0.1
```

R5

```
R5(config)# ip route 192.168.4.0 255.255.255.0 10.0.0.9
R5(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.5
```

Kiểm tra :

R5# **show ip route**

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 10.0.0.5 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C 10.0.0.4/30 is directly connected, Serial0/0/1
L 10.0.0.6/32 is directly connected, Serial0/0/1
C 10.0.0.8/30 is directly connected, Serial0/0/0
L 10.0.0.10/32 is directly connected, Serial0/0/0
S 192.168.4.0/24 [1/0] via 10.0.0.9
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.5.0/24 is directly connected, GigabitEthernet0/0
L 192.168.5.1/32 is directly connected, GigabitEthernet0/0
S* 0.0.0.0/0 [1/0] via 10.0.0.5

R6

R6(config)# ip route 192.168.4.0 255.255.255.0 10.0.0.2

R6(config)# ip route 192.168.5.0 255.255.255.0 10.0.0.6

R6(config)# ip route 0.0.0.0 0.0.0.0 200.0.0.1

R6# show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 200.0.0.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C 10.0.0.0/30 is directly connected, Serial0/0/0
L 10.0.0.1/32 is directly connected, Serial0/0/0
C 10.0.0.4/30 is directly connected, Serial0/0/1
L 10.0.0.5/32 is directly connected, Serial0/0/1
S 192.168.1.0/24 [1/0] via 192.168.6.1
S 192.168.2.0/24 [1/0] via 192.168.6.2
S 192.168.3.0/24 [1/0] via 192.168.6.3
S 192.168.4.0/24 [1/0] via 10.0.0.2
S 192.168.5.0/24 [1/0] via 10.0.0.6
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.6.0/29 is directly connected, GigabitEthernet0/0
L 192.168.6.6/32 is directly connected, GigabitEthernet0/0
200.0.0.0/24 is variably subnetted, 2 subnets, 2 masks
C 200.0.0.0/24 is directly connected, GigabitEthernet0/1
L 200.0.0.3/32 is directly connected, GigabitEthernet0/1
S* 0.0.0.0/0 [1/0] via 200.0.0.1

Bước 6 : Cấu hình NAT overload trên R6

R1
<pre>R6(config)# access-list 1 permit any R6(config)# ip nat inside source list 1 interface Gi0/1 overload R6(config)# interface Gi0/1 R6(config-if)# ip nat outside R6(config-if)# exit R6(config)# interface Gi0/0 R6(config-if)# ip nat inside R6(config-if)# exit R6(config)# interface S0/0/0 R6(config-if)# ip nat inside R6(config-if)# exit R6(config)# interface S0/0/1 R6(config-if)# ip nat inside R6(config-if)# exit</pre>

Bước 7 : Cấu hình NAT tĩnh trên router INTERNET

INTERNET
<pre>INTERNET(config)# ip nat inside source static 192.168.1.2 200.0.0.2 INTERNET(config)# interface Gi0/0 INTERNET(config-if)# ip nat inside INTERNET(config-if)# exit INTERNET(config)# interface Gi0/1 INTERNET(config-if)# ip nat outside</pre>

Bước 8 : Cấu hình Access List chặn các PC trong mạng 192.168.4.0/24 và 192.168.5.0/24 download file từ Server bằng FTP và TFTP. Những traffic còn lại được cho phép.

R6
<pre>R6(config)# access-list 100 deny tcp 192.168.4.0 0.0.0.255 host 200.0.0.2 eq 21 R6(config)# access-list 100 deny udp 192.168.4.0 0.0.0.255 host 200.0.0.2 eq 69 R6(config)# access-list 100 deny tcp 192.168.5.0 0.0.0.255 host 200.0.0.2 eq 21 R6(config)# access-list 100 deny udp 192.168.5.0 0.0.0.255 host 200.0.0.2 eq 69 R6(config)# access-list 100 permit ip any any R6(config)# interface S0/0/0 R6(config-if)# ip access-group 100 in R6(config-if)# exit R6(config)# interface S0/0/1 R6(config-if)# ip access-group 100 in</pre>